

IPAB summer internship

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This study investigated whether ensemble fusion could improve the reliability of multi-organ CT segmentation. Predictions from three pretrained segmentation models were mapped into a common label space and combined using four approaches: unweighted consensus, weighted consensus, STAPLE, and a random-forest classifier. The fused masks were evaluated against three independent expert annotations on a held-out testing set.

1 Methods

1.1 Dataset

To evaluate the proposed methods, the experiments were conducted using the CURVAS dataset [1], a multi-organ abdominal CT dataset comprising 90 scans. The dataset is divided into 20 training cases, 5 validation cases, and 65 testing cases. Each scan is accompanied by three independent expert annotations, with labels provided for the pancreas, kidneys, and liver, as summarised in Table 1. It is important to note that case UKCHLL082 was excluded from the testing set because its pancreas and liver annotations were incomplete or unsuitable for evaluation. Consequently, the final analysis was conducted on 64 testing cases.

Label	Class
0	Background
1	Pancreas
2	Kidney
3	Liver

Table 1: Label mapping used for the segmentation masks.

1.2 Constituent Models

The three segmentation models used in this study were the Universal U-Net , the Swin UNETR proposed by Tang et al., and the SuPreM SegResNet, with pretrained weights obtained from the SuPreM repository [2]. Although the models were trained using different architectures and training procedures, their training data partially overlapped, including datasets such as WORD. As shown in Table 2 and Table 3, the models also differed in their output decisions and native label spaces. The Universal U-Net and SuPreM SegResNet produced predictions in a common label space, whereas Swin UNETR produced predictions using the BTCV label convention. Consequently, label-space harmonisation was required before the model outputs could be combined using the proposed fusion methods.

CURVAS class	Universal U-Net raw channel	SegResNet raw channel	Swin UNETR raw label
Pancreas	10	10	11
Kidney	1 + 2: right and left kidney	1 + 2: right and left kidney	2 + 3: right and left kidney
Liver	5	5	6
Background	All non-target foreground labels were mapped to background	All non-target foreground labels were mapped to background	Label 0, together with all non-target labels

Table 2: Mapping from each model’s native output representation to the common CURVAS label space.

Setting	Universal U-Net	SuPreM SegResNet	Swin UNETR
Intensity range	HU clipped to $[-175, 250]$ and scaled to $[0, 1]$	HU clipped to $[-175, 250]$ and scaled to $[0, 1]$	HU clipped to $[-175, 250]$ and scaled to $[0, 1]$
Target spacing	(1.5, 1.5, 1.5) mm	(1.5, 1.5, 1.5) mm	(1.5, 1.5, 1.5) mm
ROI size	(96, 96, 96) voxels	(96, 96, 96) voxels	(96, 96, 96) voxels
Inference method	Gaussian sliding-window inference with 75% overlap	Gaussian sliding-window inference with 75% overlap	Gaussian sliding-window inference with 50% overlap
Output activation	Independent sigmoid	Independent sigmoid	Softmax followed by argmax

Table 3: Inference settings used for the three constituent segmentation models.

1.3 Fusion Methods

As discussed before, I utilised 4 techniques in order to study how ensemble methods could improve reliability.

1.3.1 Unweighted Voting

The unweighted voting approach combined predictions from all three segmentation models while giving each model equal importance. Unanimous predictions and organ labels supported by two models were retained. When two models predicted background and one predicted an organ, background was selected unless the organ prediction was connected to a region supported by stronger model agreement.. Where all three models disagreed, the label was determined using local spatial support, with the CLIP Universal U-Net prediction used as a final fallback when the disagreement remained unresolved.

1.3.2 Weighted Voting

The weighted voting approach assigned each model an organ-specific weight based on its mean Dice score over the 20 training cases. For each voxel, the weights of models predicting the same organ were combined, allowing more reliable models to contribute more strongly to the final decision. Predictions with strong weighted support were accepted directly, while weaker single-model predictions were retained only when connected to a strongly supported region of the same organ.

Model	Pancreas	Kidney	Liver
Universal U-Net	0.575394	0.868847	0.932743
SuPreM SegResNet	0.726799	0.921624	0.606787
Swin UNETR	0.361092	0.616187	0.896024
Strong Threshold	0.831643	1.203329	1.217777
Weak Threshold	0.651097	0.742517	0.751406

Table 4: Organ-specific model weights derived from mean Dice scores on the training set.

1.3.3 STAPLE

The STAPLE approach fused the three model predictions independently for each organ. Through an iterative expectation-maximization procedure, it estimated each model’s sensitivity and specificity and used these estimates to produce a probability map for each organ. A voxel was assigned to the organ with the highest probability when that probability was at least 0.5 and exceeded the second-highest probability by at least 0.1; otherwise, it was assigned to background.

1.3.4 Random Forrest Classifier (RFC)

The random-forest approach used a supervised voxel-wise classifier to learn how to combine the three model predictions. It was trained on 20 cases using annotation 1 as the target and received 15 input features: 12 one-hot encoded class predictions and one confidence value from each model. Hyperparameters were selected using patient-level five-fold cross-validation, after which a 100-tree forest with maximum depth 5 was trained on all training cases and applied to the held-out testing set.

2 Results

All composing models and fusion methods were evaluated on the same 64 held-out test cases against the three independent expert annotations. Agreement was measured using per-organ Dice, normalized surface Dice at a 1 mm tolerance, binary Cohen’s κ , and whole-map multiclass Cohen’s κ . Fleiss’ κ was calculated across the three annotators and, where applicable, the fused prediction, with a foreground-only value also reported to reduce background-class dominance. Test annotations were used solely for evaluation.

2.1 Composing Models Performance

The individual segmentation models were first evaluated to establish their agreement with the expert annotations. These results provide a baseline for assessing whether the fusion methods improve upon the composing models and the human agreement baseline. The corresponding results are presented in Table 5.

Model	Foreground Fleiss’ κ	Pancreas κ	Kidney κ	Liver κ	Multiclass Cohen’s κ
U-Net	0.779	0.595 $\pm$ 0.002	0.827 $\pm$ 0.004	0.938 $\pm$ 0.001	0.907 $\pm$ 0.000
SegResNet	0.514	0.728 $\pm$ 0.003	0.886 $\pm$ 0.004	0.625 $\pm$ 0.002	0.686 $\pm$ 0.001
Swin	0.675	0.294 $\pm$ 0.005	0.502 $\pm$ 0.006	0.911 $\pm$ 0.001	0.851 $\pm$ 0.002

Table 5: Base-model agreement metrics. Pairwise metrics are reported as mean $\pm$ standard deviation across the three model-expert comparisons.

2.2 Fusion Methods Performance

The fusion methods were evaluated using the same test cases and agreement metrics described above. Table 6 compares their performance against the human agreement baseline

Method	Foreground Fleiss’ κ	Pancreas κ	Kidney κ	Liver κ	Multiclass Cohen’s κ
Human baseline	0.879	0.907 $\pm$ 0.008	0.961 $\pm$ 0.006	0.982 $\pm$ 0.004	0.977 $\pm$ 0.004
Unweighted	0.790	0.732 $\pm$ 0.005	0.875 $\pm$ 0.005	0.931 $\pm$ 0.001	0.914 $\pm$ 0.000
Weighted	0.791	0.741 $\pm$ 0.003	0.868 $\pm$ 0.005	0.932 $\pm$ 0.001	0.915 $\pm$ 0.000
STAPLE	0.802	0.690 $\pm$ 0.005	0.896 $\pm$ 0.005	0.949 $\pm$ 0.001	0.933 $\pm$ 0.001
Random Forest	0.812	0.709 $\pm$ 0.005	0.893 $\pm$ 0.004	0.957 $\pm$ 0.002	0.938 $\pm$ 0.001

Table 6: Agreement metrics using fusion methods

Expert 1 / Expert 2	Background	Pancreas	Kidney	Liver
Background	99.953%	0.007%	0.012%	0.027%
Pancreas	5.551%	94.442%	< 0.001%	0.007%
Kidney	3.423%	0.005%	96.544%	0.028%
Liver	1.396%	0.001%	0.002%	98.601%
Mean-case multiclass Cohen's κ : 0.975				

Table 7: Mean patient-level row-normalised confusion matrix comparing Expert 1 with Expert 2.

Expert 2 / Expert 3	Background	Pancreas	Kidney	Liver
Background	99.971%	0.002%	0.008%	0.019%
Pancreas	11.530%	88.449%	0.007%	0.014%
Kidney	2.057%	< 0.001%	97.928%	0.015%
Liver	1.110%	< 0.001%	0.005%	98.885%
Mean-case multiclass Cohen's κ : 0.981				

Table 8: Mean patient-level row-normalised confusion matrix comparing Expert 2 with Expert 3.

Expert 1 / Expert 3	Background	Pancreas	Kidney	Liver
Background	99.949%	0.005%	0.013%	0.033%
Pancreas	8.371%	91.621%	0.001%	0.007%
Kidney	2.061%	< 0.001%	97.913%	0.026%
Liver	1.348%	< 0.001%	0.003%	98.649%
Mean-case multiclass Cohen's κ : 0.974				

Table 9: Mean patient-level row-normalised confusion matrix comparing Expert 1 with Expert 3.

Expert 1 / Unweighted	Background	Pancreas	Kidney	Liver
Background	99.771%	0.019%	0.051%	0.159%
Pancreas	21.347%	78.611%	0.007%	0.035%
Kidney	3.686%	0.001%	96.291%	0.023%
Liver	1.895%	0.180%	0.017%	97.909%
Mean-case multiclass Cohen's κ : 0.913				

Table 10: Mean patient-level row-normalised confusion matrix comparing Expert 1 with the unweighted fusion approach.

Expert 2 / Unweighted	Background	Pancreas	Kidney	Liver
Background	99.783%	0.016%	0.049%	0.152%
Pancreas	24.233%	75.708%	0.008%	0.051%
Kidney	4.639%	0.047%	95.289%	0.024%
Liver	2.254%	0.180%	0.020%	97.545%
Mean-case multiclass Cohen's κ : 0.914				

Table 11: Mean patient-level row-normalised confusion matrix comparing Expert 2 with the unweighted fusion approach.

Expert 3 / Unweighted	Background	Pancreas	Kidney	Liver
Background	99.786%	0.018%	0.046%	0.150%
Pancreas	22.482%	77.468%	0.006%	0.045%
Kidney	5.021%	0.001%	94.944%	0.034%
Liver	2.619%	0.181%	0.020%	97.180%
Mean-case multiclass Cohen's κ : 0.914				

Table 12: Mean patient-level row-normalised confusion matrix comparing Expert 3 with the unweighted fusion approach.

Expert 1 / Weighted	Background	Pancreas	Kidney	Liver
Background	99.782%	0.007%	0.055%	0.156%
Pancreas	31.010%	68.968%	0.014%	0.007%
Kidney	3.612%	< 0.001%	96.362%	0.026%
Liver	2.024%	0.003%	0.027%	97.946%
Mean-case multiclass Cohen's κ : 0.915				

Table 13: Mean patient-level row-normalised confusion matrix comparing Expert 1 with the weighted fusion approach.

Expert 2 / Weighted	Background	Pancreas	Kidney	Liver
Background	99.793%	0.005%	0.053%	0.149%
Pancreas	33.740%	66.222%	0.015%	0.023%
Kidney	4.600%	0.007%	95.366%	0.027%
Liver	2.392%	0.002%	0.031%	97.575%
Mean-case multiclass Cohen's κ : 0.915				

Table 14: Mean patient-level row-normalised confusion matrix comparing Expert 2 with the weighted fusion approach.

Expert 3 / Weighted	Background	Pancreas	Kidney	Liver
Background	99.796%	0.007%	0.050%	0.148%
Pancreas	31.950%	68.020%	0.013%	0.017%
Kidney	4.948%	< 0.001%	95.013%	0.039%
Liver	2.773%	0.003%	0.031%	97.193%
Mean-case multiclass Cohen's κ : 0.915				

Table 15: Mean patient-level row-normalised confusion matrix comparing Expert 3 with the weighted fusion approach.

Expert 1 / STAPLE	Background	Pancreas	Kidney	Liver
Background	99.918%	0.005%	0.022%	0.055%
Pancreas	39.457%	60.527%	0.002%	0.014%
Kidney	9.338%	0.002%	90.637%	0.024%
Liver	4.954%	0.005%	0.005%	95.036%
Mean-case multiclass Cohen's κ : 0.934				

Table 16: Mean patient-level row-normalised confusion matrix comparing Expert 1 with the STAPLE fusion approach.

Expert 2 / STAPLE	Background	Pancreas	Kidney	Liver
Background	99.926%	0.003%	0.021%	0.050%
Pancreas	42.053%	57.920%	0.003%	0.024%
Kidney	10.661%	0.008%	89.307%	0.025%
Liver	5.426%	0.004%	0.007%	94.563%
Mean-case multiclass Cohen's κ : 0.932				

Table 17: Mean patient-level row-normalised confusion matrix comparing Expert 2 with the STAPLE fusion approach.

Expert 3 / STAPLE	Background	Pancreas	Kidney	Liver
Background	99.928%	0.005%	0.018%	0.050%
Pancreas	40.407%	59.569%	0.002%	0.022%
Kidney	11.074%	0.002%	88.890%	0.034%
Liver	5.870%	0.005%	0.006%	94.119%
Mean-case multiclass Cohen's κ : 0.932				

Table 18: Mean patient-level row-normalised confusion matrix comparing Expert 3 with the STAPLE fusion approach.

Expert 1 / Random Forest	Background	Pancreas	Kidney	Liver
Background	99.916%	0.018%	0.023%	0.043%
Pancreas	23.854%	76.138%	0.007%	0.001%
Kidney	9.255%	0.009%	90.727%	0.010%
Liver	4.465%	0.063%	0.017%	95.455%
Mean-case multiclass Cohen's κ : 0.939				

Table 19: Mean patient-level row-normalised confusion matrix comparing Expert 1 with the Random Forest fusion approach.

Expert 2 / Random Forest	Background	Pancreas	Kidney	Liver
Background	99.924%	0.016%	0.021%	0.039%
Pancreas	26.615%	73.372%	0.008%	0.005%
Kidney	10.524%	0.052%	89.415%	0.010%
Liver	4.964%	0.064%	0.019%	94.953%
Mean-case multiclass Cohen's κ : 0.938				

Table 20: Mean patient-level row-normalised confusion matrix comparing Expert 2 with the Random Forest fusion approach.

Expert 3 / Random Forest	Background	Pancreas	Kidney	Liver
Background	99.925%	0.018%	0.018%	0.039%
Pancreas	24.843%	75.149%	0.005%	0.002%
Kidney	10.997%	0.009%	88.977%	0.017%
Liver	5.451%	0.065%	0.019%	94.465%
Mean-case multiclass Cohen's κ : 0.937				

Table 21: Mean patient-level row-normalised confusion matrix comparing Expert 3 with the Random Forest fusion approach.

3 Discussion

3.1 Agreement masks vs Base models

All four consensus methods improved overall agreement relative to the individual base models, although the magnitude of improvement changed by organ.

The RFC produced the strongest overall consensus, achieving the highest foreground Fleiss' k of 0.812 and the highest multiclass Cohen's κ of 0.938. Relative to the strongest base model for multiclass agreement, U-Net at 0.907, this represents an improvement of 0.031. RFC also produced the highest liver agreement, increasing liver κ from the best base-model value of 0.938 to 0.957.

STAPLE was the second-best method overall, with a multiclass κ of 0.933 and foreground Fleiss' κ of 0.802. It achieved the highest kidney agreement among the fusion methods at 0.896, slightly exceeding SegResNet's base-model kidney score of 0.886. STAPLE also improved liver agreement over all individual models, reaching 0.949.

The unweighted and weighted voting methods produced similar results. Weighted voting marginally outperformed unweighted voting in foreground agreement, pancreas agreement, liver agreement, and multiclass agreement. However, unweighted voting performed slightly better for the kidney. The differences between the two approaches were small, indicating that the organ-specific weighting strategy provided only limited additional benefit over equal voting.

The largest improvements were observed where individual models performed poorly. For example, Swin UNETR achieved a pancreas κ of only 0.294, whereas the fusion approaches obtained values between 0.690 and 0.741. Similarly, Swin's kidney agreement increased from 0.502 to approximately 0.87–0.90 under the consensus methods. The fusion approaches also substantially corrected SegResNet's comparatively weak liver performance of 0.625, producing liver κ values between 0.931 and 0.957.

However, the consensus methods did not consistently outperform the best individual model for every organ. The highest pancreas agreement among the base models was achieved by SegResNet at 0.728. Weighted voting slightly improved this to 0.741, while unweighted voting was similar at 0.732. In contrast, STAPLE and RFC obtained lower pancreas values of 0.690 and 0.709, respectively. This suggests that their stronger overall performance involved a trade-off in pancreas segmentation accuracy.

3.2 Agreement masks vs Human agreement

None of the consensus methods reached the human inter-annotator baseline. The human annotations achieved a foreground Fleiss' κ of 0.879 and a multiclass Cohen's κ of 0.977. Random Forest was the closest method, but remained lower by 0.067 for foreground Fleiss' κ and 0.039 for multiclass Cohen's κ .

The gap was largest for the pancreas. Human agreement reached 0.907, whereas the best fusion result was 0.741 from weighted voting, a difference of 0.166. The kidney and liver results were closer to the human baseline. STAPLE achieved a kidney κ of 0.896 compared with 0.961 for the experts, while Random Forest achieved a liver κ of 0.957 compared with the human value of 0.982.

Overall, the results show that consensus fusion substantially improves robustness over the constituent models, particularly by compensating for organ-specific weaknesses. Random Forest provides the strongest overall agreement, while weighted voting gives the best pancreas result and STAPLE gives the best kidney result. Nevertheless, expert-to-expert agreement remains higher across every reported metric, indicating that the fused masks do not yet match the consistency of human annotations.

References

- [1] CURVAS Challenge Organising Committee, “CURVAS Dataset,” <https://doi.org/10.5281/zenodo.13767408>, 2024.
- [2] Mr Giovanni, “SuPreM,” <https://github.com/MrGiovanni/SuPreM/tree/main>, 2024.